



Institute of Software Engineering

Diploma in Comprehensive Master Java Developer (CMJD)

Batch - CMJD106

Module – Object Oriented Programming

Assignment - 05

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01. Which of the statements below are true?

- A. Abstract classes cannot be used to instantiate objects, because they're incomplete.
- B. An abstract class must contain at least one abstract method.
- C. Constructors and static methods can be declared abstract.
- D. Abstract methods do not provide implementations.
- E. Abstract classes sometimes constitute several levels of a hierarchy.
- F. Classes that can be used to instantiate objects are called concrete classes. They provide implementations of every method they declare.
- G. A class that contains any abstract methods must be declared as an abstract class. Each concrete subclass must provide implementations of each of the superclass's abstract methods.

02. Which of the following are true statements?

- A. Any named class can be declared abstract.
- B. An incompletely implemented class must be declared abstract.
- C. An abstract class can be instantiated.
- D. An abstract class is implicitly final.
- E. An abstract class must declare at least one abstract method.
- F. An abstract class cannot extend a concrete class.

03. Which of the statements below are true?

- A. A superclass reference variable can be used to invoke only methods declared in the superclass.
- B. Operator 'instanceof' determines if an object has the 'has a' relationship with a specific type.
- C. The 'is-a' relationship applies only between the subclass and its super classes, not vice versa.
- D. Most method calls are resolved at execution time, based on the type of the object being manipulated. This process is known as dynamic binding or late binding.

04. Given Code:

```
abstract class Base{
    abstract public void myfunc();
    public void another(){
        System.out.println("Another method");
    }
}

class Abs extends Base{
    public static void main(String argv[]){
        Abs a = new Abs();
        a.amethod();
    }
}
```

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```
public void myfunc(){  
    System.out.println("My func");  
}  
public void amethod(){  
    myfunc();  
}  
}
```

What will happen when you attempt to compile and run this code?

- A. The code will compile and run, printing out the words "My Func"
- B. The compiler will complain that the Base class has non abstract methods
- C. The code will compile but complain at run time that the Base class has non abstract methods
- D. The compiler will complain that the method 'myfunc' in the base class has no body, nobody at all to lose it

05. Which of the statements below are true?

- A. An interface specifies what operations are allowed but not how they're performed.
- B. A Java interface describes a set of methods that can be called on an object.
- C. An interface cannot declare variables or constants.
- D. All objects of a class that implement multiple interfaces have the 'is-a' relationship with each implemented interface type.

06. Which of the following methods can be legally inserted at line 2

```
abstract class Customer{  
    //Line 2  
}
```

- A. void search(String name){}
- B. abstract void search(String name){}
- C. abstract void search(String name);
- D. private abstract void search(String name);
- E. static abstract void search(String name);
- F. private static abstract void search(String name);
- G. public abstract void search(String name);

07. Compare and contrast abstract classes and interfaces. Why would you use an abstract class? Why would you use an interface?

08. Given:

```
interface A {  
    void m1();  
    // 1
```

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```
        public void m2();           // 2
        protected void m3();       // 3
        private void m4();         // 4
    }
```

What is the result of attempting to compile the code?

- A. Compiler error at line 1.
- B. Compiler error at line 2.
- C. Compiler error at line 3
- D. Compiler error at line 4.

09. Which of the following statements are true?

- A. All of the variables in an interface are implicitly static .
- B. All of the variables in an interface are implicitly final .
- C. All of the methods in an interface are implicitly abstract.
- D. A method in an interface can access class level variables .

10. Which of the following are legal declarations for nonnested classes and interfaces?

- A. final abstract class Test
- B. public static interface Test{}
- C. final public class Test{}
- D. protected abstract class Test{}
- E. protected interface Test{}
- F. abstract public class Test{}

11. Which of the following statements are true?

- A. An interface can only contain methods and not variables .
- B. Interfaces cannot have constructors.
- C. class may extend only one other class and implement only one interface.
- D. Interfaces are the Java approach to addressing its lack of multiple inheritance, but require.
- E. implementing classes to create the functionality of the Interfaces.

12. Given Code:

```
interface I10 {
    String name = "I10";
    String s10 = "I10.s10";
}
interface I20 {
    String name = "I20";
    String s20 = "I20.s20";
}
```

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```
class C20 implements I10, I20 {  
  
    public static void main(String[] args) {  
        System.out.print(I10.s10+",");  
        System.out.print(I20.s20+",");  
        System.out.print(I20.name);  
    }  
}
```

What is the result of attempting to compile and run the program?

- A. Prints: I10.s10,I20.s20,I10
- B. Prints: I10.s10,I20.s20,I20
- C. Prints: I10.s10,I20.s20,
- D. Prints: I10.s10,I20.s20,null
- E. Compiler error at line 9.
- F. Compiler error at line 11.
- G. Compiler error at line 12.
- H. Compiler error at line 13.